

$^{35}\text{Cl}(\gamma, \gamma')$ [1966Ho02](#), [1962Bo17](#), [1991Li12](#)

[1966Ho02](#): a bremsstrahlung photon beam was produced by impinging 1.4-MeV electrons on a thin layer of lead embedded in a water-cooled aluminium disc at the Diamond Research Laboratory in Johannesburg. The target was C_2Cl_6 . Scattered γ rays were detected using a 7.6-cm by 7.6-cm NaI crystal. Deduced the width for the 1.22-MeV first excited state using the self-absorption method of nuclear resonance fluorescence.

[1962Bo17](#): an electron beam was produced from the MIT Van de Graaff generator, passed through a 76- μm aluminum window and a 25- μm platinum foil, and stopped in the cooling water of the electron target. γ rays were detected using a 5.1-cm by 5.1-cm NaI crystal. Lifetimes of ^{35}Cl levels are deduced from the measured intensity of the resonance fluorescence radiation which occurs when a nucleus is bombarded by bremsstrahlung.

[1991Li12](#): a bremsstrahlung photon beam was produced by focusing a 4.1-MeV electron beam onto a radiator target of a gold disc on a water-cooled copper backing at the bremsstrahlung facility at the Stuttgart Dynamitron Accelerator. The target was a stack of discs that contain ^{13}C , ^{23}Na , ^{27}Al , ^{31}P , ^{35}Cl , ^{48}Ti . γ rays were detected using 3 Ge detectors. Deduced σ of photon scattering.

Other: [1972Sh07](#).

 ^{35}Cl Levels

$\Gamma_0\Gamma_f/\Gamma$ from [1991Li12](#). Γ_0 , Γ_f , and Γ are the decay width to the ground state, the decay width to the final state, and the total width, respectively.

<u>E(level)[†]</u>	<u>T_{1/2}</u>	<u>Comments</u>
0		
1220	90 fs +21-14	T _{1/2} : from lifetime=0.13 ps +3-2 in 1966Ho02 with resonance fluorescence. Other: lifetime=0.30 ps 10 in 1962Bo17 with resonance fluorescence assuming J=1/2.
1760	0.42 ps 15	T _{1/2} : from lifetime=0.60 ps 21 in 1962Bo17 with resonance fluorescence assuming J=5/2.
2693.7		$\Gamma_0\Gamma_f/\Gamma=12$ meV 3 (1991Li12).
3002.7		$\Gamma_0\Gamma_f/\Gamma=26$ meV 11 (1991Li12).
3918.6		$\Gamma_0\Gamma_f/\Gamma=63$ meV 18 (1991Li12).

[†] From E γ data.

 $\gamma(^{35}\text{Cl})$

<u>Eγ</u>	<u>E_i(level)</u>	<u>E_f</u>	<u>Comments</u>
1220	1220	0	E γ : from 1962Bo17 and 1966Ho02 .
1760	1760	0	E γ : from 1962Bo17 .
2693.6 [†]	2693.7	0	
3002.6 [†]	3002.7	0	
3918.4 [†]	3918.6	0	

[†] From [1991Li12](#), which quoted from [1978En02](#) evaluation.

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Level Scheme

